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An electrolyte- and catalyst-free electrooxidative sulfonylation of imidazo[1,2-*a*]pyridines†

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Direct electrochemical oxidation has been widely applied in organic synthesis due to its simple and mild reaction conditions, high selectivity, few by-products, high product purity and high productivity. Up to now, the efficient synthetic methods for 3-sulfonylated imidazo[1,2-*a*]pyridines have been the C–H activation of imidazo[1,2-*a*]pyridines with a photosensitizer or transition-metal-catalyst at a high reaction temperature. In this article, an electrolyte- and catalyst-free electrooxidative approach to produce sulfonylated imidazo[1,2-*a*]pyridines was investigated. Various imidazo[1,2-*a*]pyridines and sodium sulfinates could be selectively transformed into the 3-sulfonylated imidazo[1,2-*a*]pyridines in good to excellent yields. Based on the results of control experiments and theoretical calculations, we proposed that the studied reaction might occur *via* a radical process. The spectral properties of imidazo[1,2-*a*]pyridines might be controlled by the modification of their 8-position.

Introduction

Electrosynthesis has long been recognized as an environmentally friendly synthetic tool in organic synthesis owing to the application of electrons as inexpensive, mass-free, renewable and inherently safe redox reagents.[1] Direct electrooxidation provides an appealing alternative to traditional oxidation reactions. In the direct electrooxidation process, sacrificial reagents (such as exogenous-catalyst and/or oxidant) can be successfully avoided, and only electrical energy is required.[2] As is widely known, supporting electrolyte additives are necessary to ensure sufficient ionic conductivity for most organic electrosyntheses.[3] However, an excess amount of electrolyte leads to a tedious workup and the generation of additional waste. To solve the above problems, electrolyte-free electrooxidation is a better choice. Although several electrooxidation reactions involving electrolyte-free conditions have been developed to construct C–C,[4] C–O[5] and C–S[6] bonds, electrolyte-free electrooxidation is still relatively rare.

Imidazo[1,2-*a*]pyridines, as a class of nitrogen-containing heterocycles, are widely found in many biologically active molecules and pharmaceutical intermediates.[7] In general, they are applied into $GABA_A$ receptor agonists, proton pump inhibitors, aromatase inhibitors, NSAIDs and other classes of drugs.[8] Imidazo[1,2-*a*]pyridine derivatives also exhibit antibacterial, antifungal, anti-inflammatory, antitumor, and antiviral properties.[9] Several drug formulations containing imidazo[1,2-*a*]pyridines are available on the market, such as alpidem,[10] zolpidem,[11] olprinone,[12] and minodronic acid.[13] In the field of materials science, imidazo[1,2-*a*]pyridines have been proven to be useful in luminescent materials and fluorescent probes.[14]

Due to these attractive applications, various functionalization methods of imidazo[1,2-*a*]pyridines have become a hot research topic in recent years.[15] Among them, the synthesis of 3-sulfonylated imidazo[1,2-*a*]pyridines has only been realized *via* the three following pathways: (1) the oxidation of sulfenylimidazo[1,2-*a*]pyridines in the presence of *m*-CPBA;[16] (2) Cu_2O-catalyzed[17] or I_2-mediated[18] radical reactions of 2-aryl imidazo[1,2-*a*]pyridines at a high reaction temperature; (3) visible-light-induced three component coupling reaction with an organic photocatalyst (Scheme 1).[19] However, the usage of transition-metal catalysts, chemical oxidants, free radical initiators and photosensitizers not only affects the purity of the product, but also is not conducive to environmental protection. Therefore, the development of a simple, efficient, green and direct approach is imperative for the production of 3-sulfonylated imidazo[1,2-*a*]pyridines. With the conception of direct electrooxidation, herein we demonstrate an electrolyte- and catalyst-free electrooxidative sulfonylation to access 3-sulfonylated imidazo[1,2-*a*]pyridines. The mechanism will be investigated by experimental and density functional theory (DFT) approaches.

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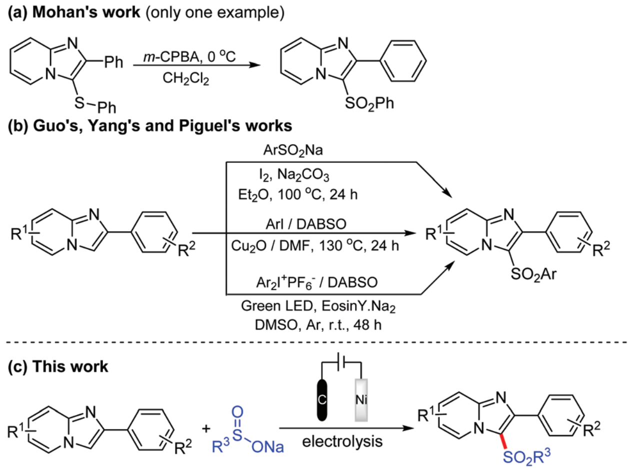


Scheme 1 Synthetic methods for the sulfonylation of imidazopyridines.

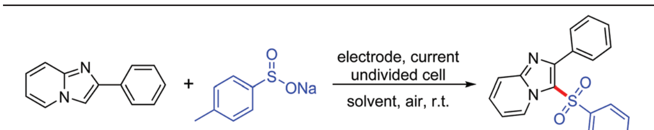
Results and discussion

Inspired by previous work,[20] the reaction of 2-phenylimidazo[1,2-*a*]pyridine (**1a**) with sodium *p*-toluenesulfinate (**2a**) was investigated as a model reaction. Fortunately, the model reaction proceeded to afford the target product 2-phenyl-3-tosylimidazo[1,2-*a*]pyridine (**3aa**) in moderate yield (52%) at a constant current density of 5 mA cm^{-2} in a C|C undivided cell without any electrolyte (Table 1, entry 1). The product yield was not improved by changing the current density (Table 1, entries 2 and 3). It was found that a Pt or Ni electrode was the best choice for this transformation (Table 1, entries 4–8). Other solvents such as dimethylformamide (DMF), ethanol (EtOH) and tetrahydrofuran (THF) reduced the yield of compounds **3aa** (Table 1, entries 9–11). The conversion yield was increased with increased water content (Table 1, entries 13 and 14). In addition, the optimal reaction time was 9 h (entries 13 and 15, 16). Although the product yield of **3aa** could be improved under an Ar atmosphere (Table 1, entry 17), the simpler reaction condition (under air) would be more convenient and accessible for further investigation. During the reaction process, charge consumption was 5.6 F mol^{-1}. Because two equivalents of electrons were required for one equivalent of **3aa**, the faradaic efficiency (FE) was 36%.[21]

With the optimal reaction conditions in hand, the scope of 2-arylimidazo[1,2-*a*]pyridines (**1**) and sodium sulfinates (**2**) was explored (Scheme 2). Initially, sodium arylsulfinates (**2**) with both electron-donating and electron-withdrawing groups were found have a similar activity (**2a–2g**), except for sodium 4-chlorobenzenesulfinate (**2f**). The products **3aa–3ag** were isolated in good yields. Due to the higher second oxidation potential of sodium naphthalene-2-sulfinate (**2h**) (see ESI, Fig. S1†), the target product (**3ah**) was only detected in a trace amount. Surprisingly, sodium heteroaryl sulfinate (**2i–2j**) and sodium alkyl sulfinate (**2k**) were suitable for the electrocatalytic method to give the corresponding products **3ai–3ak** in good to excellent yields. The activities of 2-phenylimidazo[1,2-*a*]pyridines linked with electron-donation on C6, C7, or C8 positions were higher than those linked with electron-withdrawing groups (**3ba–3ma**). It is worth mentioning that electrocatalytic debromination was found as a by-product (**3aa**, 40%) with the formation of compounds **3ma**. It might be the main reason for the low yield of product **3ma**. In addition, a similar electronic effect was also observed for 2-arylimidazo[1,2-*a*]pyridines (**1n–1r**) bearing various functional groups. The excellent yields (**3na–3pa**) were detected from the reaction involving the 2-arylimidazo[1,2-*a*]pyridines with electron-donating groups (**1n–1m**) and a weak electron-withdrawing group (**1p**). The structures of **3aa**, **3ad**, **3ca**, **3da** and **3na** were determined by X-ray diffractometry (see the ESI†).

In order to gain insight into the mechanism of the method, several control experiments were carried out (Scheme 3). Initially, an excess amount of 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO) as a free radical scavenger could completely quench this reaction. Although the sulfonylation was not completely quenched by butylated hydroxytoluene (BHT), three strong molecular ion peaks were detected by ESI-MS: $[\mathbf{I} + Na]^+$ (exact mass: 397.1806), $[\mathbf{II} + H]^+$ (exact mass 413.2590) and $[\mathbf{III}]^+$ (exact mass 155.0161) (exp. 1–3 and see ESI†). The scale-up reaction with 1 mmol **1a** gave product **3aa** with an excellent yield (exp. 4). Sodium *p*-toluenesulfinate (**2a**) and 2-phenylimidazo[1,2-*a*]pyridine (**1a**) were oxidized at 0.93 V and 1.55 V *vs.* Ag/AgCl, respectively (Fig. 1A). The lower oxidation potential of *p*-toluenesulfinate (**2a**) makes it easy to be firstly oxidized at the anode. Based on these results, a control experiment was conducted at the corresponding anode potential (0.96 V *vs.* Ag/AgCl). The desired product **3aa** was only isolated in 22%

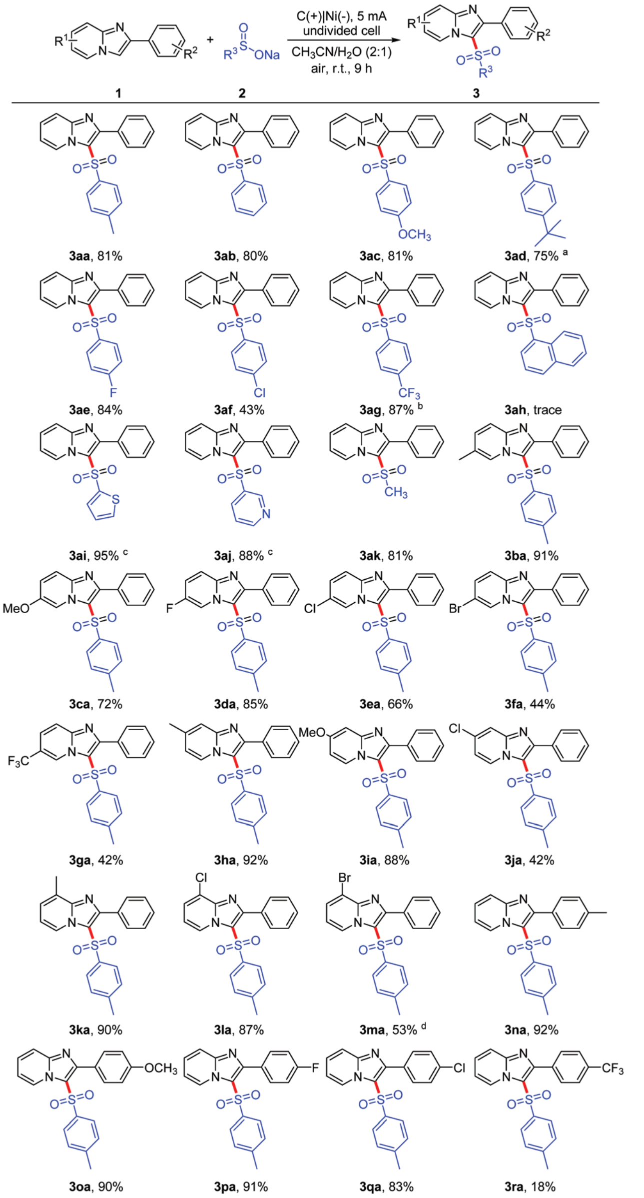
Table 1 Optimization of the reaction conditions[a]



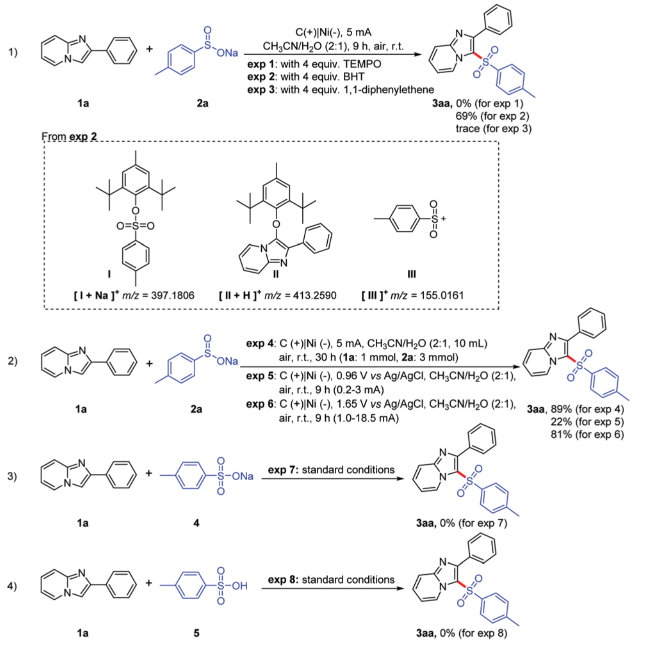
1a **2a** **3aa**

Entry	Anode|cathode	Solvent	Yield[b] (%)
1	C|C	CH_3CN/H_2O (6 : 1)	52
2[c]	C|C	CH_3CN/H_2O (6 : 1)	50
3[d]	C|C	CH_3CN/H_2O (6 : 1)	45
4	C|Pt	CH_3CN/H_2O (6 : 1)	75
5	C|Ni	CH_3CN/H_2O (6 : 1)	75
6	C|Zn	CH_3CN/H_2O (6 : 1)	68
7	C|Fe	CH_3CN/H_2O (6 : 1)	70
8	C|Cu	CH_3CN/H_2O (6 : 1)	62
9	C|Ni	DMF/H_2O (6 : 1)	42
10	C|Ni	EtOH/H_2O (6 : 1)	N.R.
11	C|Ni	THF/H_2O (6 : 1)	46
12	C|Ni	CH_3CN/H_2O (4 : 1)	79
13	**C|Ni**	**CH_3CN/H_2O (2 : 1)**	**81**
14	C|Ni	CH_3CN/H_2O (1 : 1)	14
15[e]	C|Ni	CH_3CN/H_2O (2 : 1)	72
16[f]	C|Ni	CH_3CN/H_2O (2 : 1)	73
17[g]	C|Ni	CH_3CN/H_2O (2 : 1)	84

[a] Reaction conditions: **1a** (0.3 mmol), **2a** (3 equiv.), solvent (7 mL), constant current = 5 mA, undivided cell, under air at room temperature for 9 h. [b] Isolated yields. [c] 3 mA. [d] 7 mA. [e] 8 h. [f] 10 h. [g] Under argon.



Scheme 2 The substrate scope of 2-arylimidazo[1,2-*a*]pyridines and sodium sulfinates. Reaction conditions: **1** (0.3 mmol), **2** (3 equiv.), CH_3CN/H_2O (7 mL), constant current = 5 mA, undivided cell, under air at room temperature for 9 h, isolated yields. [a]6 h. [b]7 h. [c]8 h. [d]**3aa** as a by-product was isolated in 40% yield.



Scheme 3 Control experiments.

yield (exp. 5) due to the low current density (0.2–3 mA cm^{-2}). The applied potential of E = 1.65 V *vs.* Ag/AgCl caused an overall increase of the cell current up to 18.5 mA cm^{-2}, and therefore the desired product **3aa** was isolated in 81% yield (exp. 6). This indicated that the oxidation of compound **1a** on the anode was the key process to realize this transformation, and the electro-oxidation of substrate **2a** on the anode might improve the reaction. On the other hand, the influence of the potential scan rate on the electrochemical responses of 2-phenylimidazo[1,2-*a*]pyridine (**1a**) and sodium *p*-toluenesulfinate (**2a**) in CH_3CN was studied by cyclic voltammetry. With an increase of scan rate, the oxidation peak currents increased linearly with their scan rate (Fig. 1B, C and E, F). The linear regression equations could be calculated as follows: I (μA) = 0.36554ν (mV s^{-1}) + 112.26168 (R = 0.96161) for compound **1a** (Fig. 1C) and I (μA) = 0.10563ν (mV s^{-1}) + 28.4851 (R = 0.97767) for compound **2a** (Fig. 1F). In addition, the E_{pa} was also positively correlated with $\ln\nu$, the corresponding linear regression equations were E_{pa} = 0.04124$\ln\nu$ + 1.34822 (R = 0.98284) for compound **1a** (Fig. 1D) and E_{pa} = 0.0261$\ln\nu$ + 0.82034 (R = 0.96469) for compound **2a** (Fig. 1G). For an absorption-controlled and irreversible electrode process, the number of electrons (n) was calculated to be 2 for sodium *p*-toluenesulfinate (**2a**) and 1 for 2-phenylimidazo[1,2-*a*]pyridine (**1a**) by the Laviron method.[22] When *p*-toluenesulfonic acid or sodium *p*-toluenesulfonate was used as the starting material to replace compound **2a**, no desired product **3aa** was isolated (exp. 7 and 8). In addition, adduct **II** was formed from the reaction between BHT with the 2-phenylimidazo[1,2-*a*]pyridine radical which might be afforded *via* the electrooxidation of compound **1a** with the accompanying H^+ elimination. These results proved that: (1) the electrooxidation might proceed *via* a radial process; (2) the oxidation of compounds **1a** and **2a** was an absorption-controlled process; (3) the electrooxidation step included two steps of a two-electron process for compound **2a**, and one step of a one-electron process for compound **1a**. However, the aryl-sulfonyl cation produced at the anode might not be involved in the sulfonylation. It might only act as an addition electrolyte.

To further verify the reaction mechanism and simplify the calculation process, the reaction pathway was examined by DFT at the M06-2X and def2-TZVP(D) standard basis set level of theory using 2-phenylimidazo[1,2-*a*]pyridine (**1a**) and

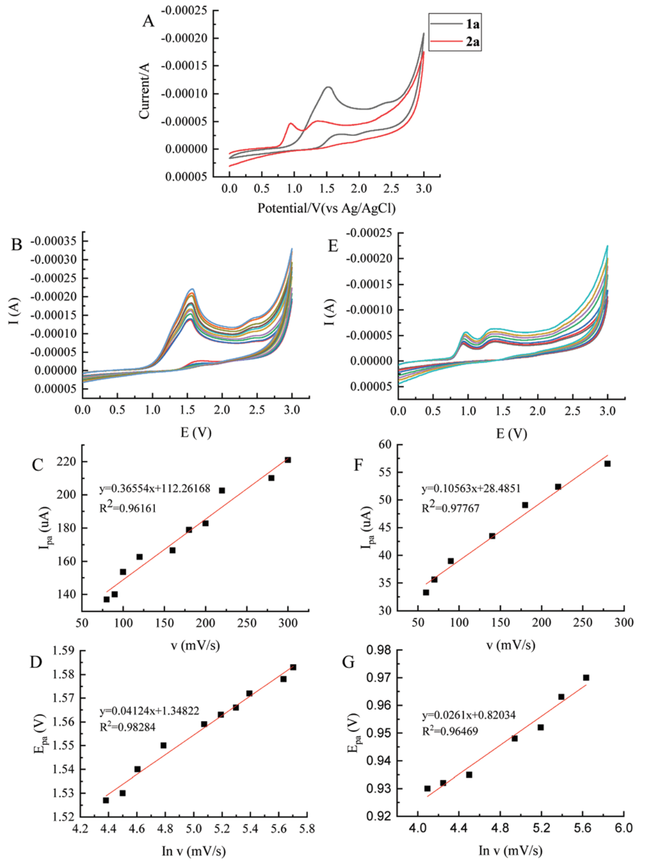


Fig. 1 (A) Cyclic voltammograms of **1a** and **2a** in CH_3CN at 100 mV s^{-1} scan rates. (B) Cyclic voltammograms of **1a** in CH_3CN at different scan rates. Curves were obtained at 80, 90, 100, 120, 160, 180, 200, 220, 280 and 300 mV s^{-1}, respectively. (C) The plot of the peak current *vs.* scan rate. (D) The relationship between E_{pa} and $\ln \nu$. (E) Cyclic voltammograms of **2a** in CH_3CN at different scan rates. Curves were obtained at 60, 70, 90, 140, 180, 220 and 280 mV s^{-1}, respectively. (F) The plot of the peak current *vs.* scan rate. (G) The relationship between E_{pa} and $\ln \nu$.

sodium methanesulfinate (**2k**) as the starting materials. The free energy barriers in the rate determining addition step for C–S bond formation and the energy profiles of the reaction are demonstrated in Fig. 2. Based on the control experiment results, we believed that the optimized reaction conditions are sufficient to enable the electrooxidation of 2-phenylimidazo[1,2-*a*]pyridine (**1a**) and sodium methanesulfinate (**2k**) to afford the methylsulfonyl radical (**B**, 115.13 kcal mol^{-1}), the methylsulfonyl cation (**C**, 145.26 kcal mol^{-1}) and the 2-phenylimidazo[1,2-*a*]pyridine cation radical **G** (137.11 kcal mol^{-1}). Subsequently, there are three possible pathways to generate the target product: (1) the radical addition could not be carried out spontaneously ($\Delta G > 0$), the energy barrier of the nucleophilic attack by radical **B** on the C3-position of 2-phenylimidazo[1,2-*a*]pyridine (**1a**) is not high (12.18 kcal mol^{-1}). The oxidation from radical **E** to cation **F** might also occur at the anode (94.96 kcal mol^{-1}) (2) due to the high acidity (pK_a = 17.13) of the 2-phenylimidazo[1,2-*a*]pyridine cation radical (**G**), it easily undergoes dehydrogenation to generate radical **H** ($\Delta G = -18.51$ kcal mol^{-1}) with the base generated *via* electrolysis at the cathode. The target product **3ak** could be isolated from the reaction between the two radicals **B** and **H** ($\Delta G = -68.49$ kcal mol^{-1}) (3) methylsulfonyl cation (**C**) reacts with 2-phenylimidazo[1,2-*a*]pyridine (**1a**) to afford **D** because of the mutual electrostatic attraction (Fig. 3).[23] However, the energy required for a 1,2-migration from the N1 to C2-position, 42.88 kcal mol^{-1}, is very large indeed. Therefore, this process might not be realized using our optimized reaction conditions.

Based on the results of the above experimental and theoretical studies, a possible reaction mechanism is demonstrated (Scheme 4). Initially, sodium sulfinates (**2**) and 2-arylimidazo[1,2-*a*]pyridines (**1**) adsorbed on the anode are oxidated to afford the corresponding active intermediates sulfonyl radical (**B**) and cation radical (**G**). The sulfonyl radical (**B**) could be

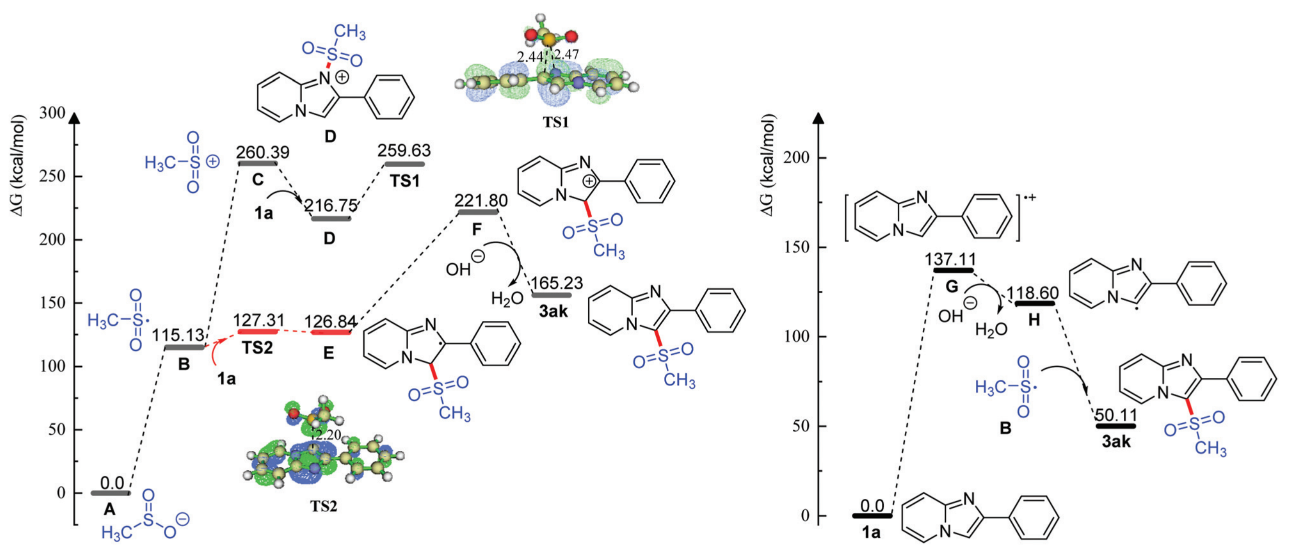


Fig. 2 DFT calculations. Computed Gibbs free energy profile for the C–H functionalization of **1a** with **2k**, and spin density of the transition states TS. Energies are given in kcal mol^{-1}. Calculated by the Gaussian 16 program using M06-2X/def2-TZVP(D)/CH_3CN : H_2O (PCM).[24]

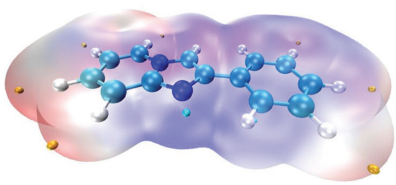
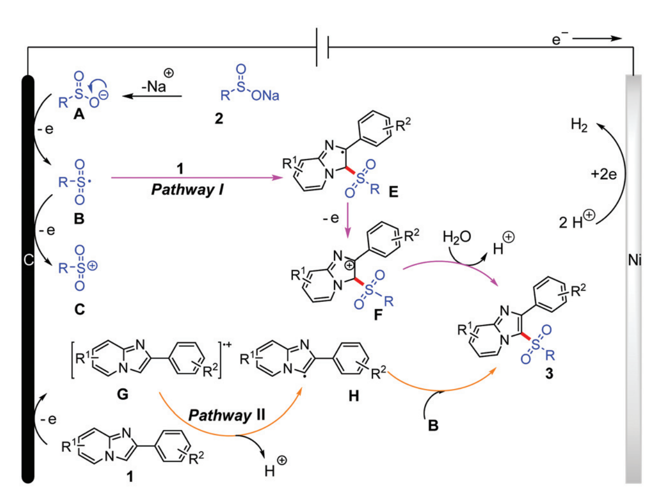


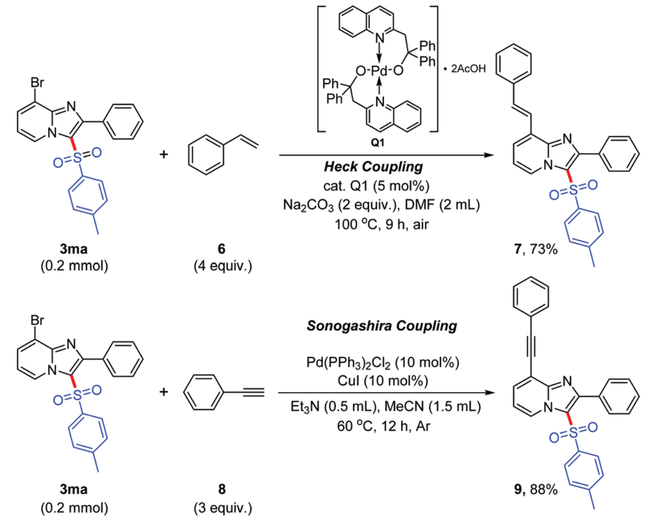
Fig. 3 Electrostatic potential (ESP)[25] of **1a**. The ESP mapping was carried out using the Multiwfn Version 3.7(dev)[26] and VMD version 1.9.3 programs.[27]



Scheme 4 The proposed mechanism.

transformed into the sulfonyl cation (**C**) by further oxidation on the anode. Subsequently, the sulfonation might proceed *via* two pathways. Pathway I: the sulfonyl radical (**B**) reacts with 2-arylimidazo[1,2-*a*]pyridines (**1**) to form radical **E**. Through the electrooxidation, radical **E** is also transformed into cation (**F**). With the help of water, cation (**F**) undergoes aromatization to produce product **3**. Pathway II: radical (**H**) is produced *via* the deprotonation of the 2-arylimidazo[1,2-*a*]pyridine cation radical (**G**). The target product **3** is produced from the radical coupling reaction of the sulfonyl radical (**B**) with the 2-arylimidazo[1,2-*a*]pyridine radical (**H**).

Due to the excellent spectral properties of imidazo[1,2-*a*]pyridines,[14,28] we further studied the spectral properties of the synthesized sulfonylated imidazo[1,2-*a*]pyridines. With the help of Pd-catalyzed Heck and Sonogashira coupling reactions, the 8-aryl vinylation and arylethynylation of imidazo[1,2-*a*]pyridine efficiently synthesized (**7**) and (**9**), respectively, and their unique spectral properties are shown in Scheme 5.[29] The results of the spectral analysis show that the UV absorption and fluorescence emission of the sulfonated product (**3aa**) are blue shifted compared with those of the non-sulfonated 2-phenylimidazo[1,2-*a*]pyridine (**1a**). When the arylvinyl or arylethynyl group is introduced into the 8-position of imidazo[1,2-*a*]pyridine (**7** and **9**), the UV absorption and fluorescence emission of the sulfonated products are significantly red shifted (Fig. 4 and Table 2). The preliminary results indicate that the modification of the 8-position of imidazo[1,2-*a*]pyridines is a feasible way to realize the control of their spectral properties.



Scheme 5 The extended experiments of **3ma**.

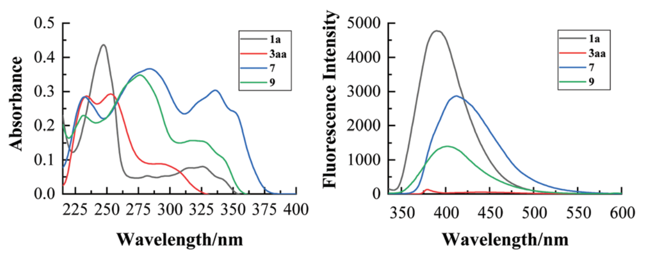


Fig. 4 UV and FL spectra of compounds **1a**, **3aa**, **7** and **9**.

Table 2 Absorption, emission and fluorescence quantum yields (Φ_F) of compounds **1a**, **3aa**, **7** and **9**[a,b]

	λ_{abs} (nm)	λ_{em} (nm)	Φ_F
1a	247, 326	389	0.43
3aa	234, 253, 291	440	0.006
7	232, 284, 336	411	0.16
9	231, 276, 317	402	0.08

[a] UV and FL spectra were recorded at a concentration of 1×10^{-5} M in CH_3CN. [b] The fluorescence quantum yield was calculated by using quinine sulfate as an external standard.

Conclusions

In conclusion, we have developed an electrolyte- and catalyst-free electrooxidative approach to produce sulfonylated imidazo[1,2-*a*]pyridines. This method has the advantages of simple and mild reaction conditions, a high selectivity, few by-products, high product purity and high productivity. The optimized reaction conditions are suitable for various 2-arylimidazo[1,2-*a*]pyridines and sodium sulfinates. The results of the control experiments and theoretical calculations indicated that the sulfonylation could proceed *via* the radical addition of a sulfonyl radical with imidazo[1,2-*a*]pyridines and the radical coupling reaction between a sulfonyl radical and imidazo[1,2-*a*]pyridines. Although 3-sulfonyl substitution of imidazo[1,2-*a*] pyridines reduced the fluorescence emission, the spectral properties might be controlled through the modification of their 8-position.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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Notes and references

1 (*a*) A. Wiebe, T. Gieshoff, S. Möhle, E. Rodrigo, M. Zirbes and S. R. Waldvogel, Electrifying Organic Synthesis, *Angew. Chem., Int. Ed.*, 2018, **57**, 5594–5619; (*b*) S. Tang, Y. Liu and A. Lei, Electrochemical Oxidative Cross-coupling with Hydrogen Evolution: A Green and Sustainable Way for Bond Formation, *Chem*, 2018, **4**, 27–45.

2 (*a*) D. Wang, P. Wang, S. Wang, Y.-H. Chen, H. Zhang and A. Lei, Direct electrochemical oxidation of alcohols with hydrogen evolution in continuous-flow reactor, *Nat. Commun.*, 2019, **10**, 2796; (*b*) Y. Yuan and A. Lei, Electrochemical Oxidative Cross-Coupling with Hydrogen Evolution Reactions, *Acc. Chem. Res.*, 2019, **52**, 3309–3324.

3 (*a*) M. Li, C. Wang, Z. Chen, K. Xu and J. Lu, New Concepts in Electrolytes, *Chem. Rev.*, 2020, **120**, 6783–6819; (*b*) M. König, J. Vaes, E. Klemm and D. Pant, Solvents and Supporting Electrolytes in the Electrocatalytic Reduction of CO_2, *iScience*, 2019, **19**, 135–160; (*c*) C. Zhang, Z. Zhang, Z. He and D. Fu, New insights into the relationship between anode material, supporting electrolyte and applied current density in anodic oxidation processes, *Electrochim. Acta*, 2017, **229**, 55–64.

4 J. L. Röckl, D. Schollmeyer, R. Franke and S. R. Waldvogel, Dehydrogenative Anodic C-C Coupling of Phenols Bearing Electron Withdrawing Groups, *Angew. Chem.*, 2020, **59**, 315–319.

5 R. Horcajada, M. Okajima, S. Suga and J.-i. Yoshida, Microflow electroorganic synthesis without supporting electrolyte, *Chem. Commun.*, 2005, **10**, 1303–1305.

6 J. Nikl, S. Lips, D. Schollmeyer, R. Franke and S. R. Waldvogel, Direct Metal- and Reagent-Free Sulfonylation of Phenols with Sodium Sulfinates by Electrosynthesis, *Chem. – Eur. J.*, 2019, **25**, 6891–6895.

7 (*a*) K. S. Gudmundsson, S. D. Boggs, J. G. Catalano, A. Svolto, A. Spaltenstein, M. Thomson, P. Wheelan and S. Jenkinson, Imidazopyridine-5,6,7,8-tetrahydro-8-quinolinamine derivatives with potent activity against HIV-1, *Bioorg. Med. Chem. Lett.*, 2009, **19**, 6399–6403; (*b*) A. Gallud, O. Vaillant, L. T. Maillard, D. P. Arama, J. Dubois, M. Maynadier, V. Lisowski, M. Garcia, J. Martinez and N. Masurier, Imidazopyridine-fused [1,3]-diazepinones: Synthesis and antiproliferative activity, *Eur. J. Med. Chem.*, 2014, **75**, 382–390; (*c*) L. Dyminska, Imidazopyridines as a source of biological activity and their pharmacological potentials-Infrared and Raman spectroscopic evidence of their content in pharmaceuticals and plant materials, *Bioorg. Med. Chem.*, 2015, **23**, 6087–6099.

8 (*a*) K. A. Wafford, M. B. vanNiel, Q. P. Ma, E. Horridge, M. B. Herd, D. R. Peden, D. Belelli and J. J. Lambert, Novel compounds selectively enhance δ subunit containing $GABA_A$ receptors and increase tonic currents in thalamus, *Neuropsychobiology*, 2009, **56**, 182–189; (*b*) N. Takahashi, T. Terao, T. Oga and M. Okada, Comparison of Risperidone and Mosapramine Addition to Neuroleptic Treatment in Chronic Schizophrenia, *Neuropsychobiology*, 1999, **39**, 81–85; (*c*) S. Feng, D. Hong, B. Wang, X. Zheng, K. Miao, L. Wang, H. Yun, L. Gao, S. Zhao and H. C. Shen, Discovery of Imidazopyridine Derivatives as Highly Potent Respiratory Syncytial Virus Fusion Inhibitors, *ACS Med. Chem. Lett.*, 2015, **6**, 359–362.

9 (*a*) N. M. Shukla, D. B. Salunke, E. Yoo, C. A. Mutz, R. Balakrishna and S. A. David, Antibacterial activities of Groebke-Blackburn-Bienaymé-derived imidazo[1,2-*a*] pyridin-3-amines, *Bioorg. Med. Chem.*, 2012, **20**, 5850–5863; (*b*) C. Hamdouchi, J. D. Blas, M. D. Prado, J. J. Gruber, B. A. Heinz and L. J. Vance, 2-Amino-3-substituted-6-[(*E*)-1-phenyl-2-(*N*-methylcarbamoyl)vinyl]imidazo-[1,2-*a*]pyridines as a Novel Class of Inhibitors of Human Rhinovirus: Stereospecific Synthesis and Antiviral Activity, *J. Med. Chem.*, 1999, **42**, 50–59; (*c*) Y. Rival, G. Grassy, A. Taudou and R. Ecalle, Antifungal activity in vitro of some imidazo [1,2-*a*]pyrimidine derivatives, *Eur. J. Med. Chem.*, 1991, **26**, 13–18.

10 B. Musch, P. L. Morselli and P. Priore, Clinical studies with the new anxiolytic alpidem in anxious patients: An overview of the European experiences, *Pharmacol., Biochem. Behav.*, 1988, **29**, 803–806.

11 T. S. Harrison and G. M. Keating, Zolpidem A Review of Its Use in the Management of Insomnia, *CNS Drugs*, 2005, **19**, 65–89.

12 K. Mizushige, T. Ueda, K. Yukiiri and H. Suzuki, Olprinone: A Phosphodiesterase III Inhibitor with Positive Inotropic and VaSodilator Effects, *Cardiovasc. Drug Rev.*, 2002, **20**, 163–174.

13 L. A. Sorbera, J. Castaner and P. A. Leeson, Minodronic Acid, *Drugs Future*, 2002, **27**, 935–941.

14 (*a*) J. Y. Lee, J. Y. Shim, H. K. Kim, D. Ko, M.-H. Baik and E. J. Yoo, 3,5-Diarylimidazo[1,2-*a*]pyridines as Color-Tunable Fluorophores, *J. Org. Chem.*, 2017, **82**, 4352–4361; (*b*) N. Shao, G.-X. Pang, C.-X. Yan, G.-F. Shi and Y. Cheng, Reaction of β-Lactam Carbenes with 2-Pyridyl Isonitriles: A One-Pot Synthesis of 2-Carbonyl-3-(pyridylamino)imidazo [1,2-*a*]pyridines Useful as Fluorescent Probes for Mercury Ion, *J. Org. Chem.*, 2011, **76**, 7458–7465.

15 (*a*) J. Koubachi, S. E. Kazzouli, M. Bousmina and G. Guillaumet, Functionalization of Imidazo[1,2-*a*]pyridines by Means of Metal-Catalyzed Cross-Coupling Reactions, *Eur. J. Org. Chem.*, 2014, **2014**, 5119–5138; (*b*) A. K. Bagdi and A. Hajra, Visible light promoted C-H functionalization of Imidazoheterocycles, *Org. Biomol. Chem.*, 2020, **18**, 2611–2631; (*c*) O. P. S. Patel, N. K. Nandwana, L. J. Legoabe, B. C. Das and A. Kumar, Recent Advances in Radical C-H Bond Functionalization of Imidazoheterocycles, *Adv. Synth. Catal.*, 2020, **362**, 4226–4255.

16 D. C. Mohan, M. S. Rao, C. Ravi and S. Adimurthy, Copper(I) Iodide Catalyzed Aerobic Oxidative C–N and C–S bond formations through C–H Activation: Synthesis of Functionalized Imidazo[1,2-*a*]pyridines, *Asian J. Org. Chem.*, 2014, **3**, 609–613.

17 D. Yang, P. Sun, W. Wei, F. Liu, H. Zhang and H. Wang, Copper-Catalyzed Regioselective Cleavage of C-X and C-H Bonds: A Strategy for Sulfur Dioxide Fixation, *Chem. – Eur. J.*, 2018, **24**, 4423–4427.

18 Y.-J. Guo, S. Lu, L.-L. Tian, E.-L. Huang, X.-Q. Hao, X.-J. Zhu, T. Shao and M.-P. Song, Iodine-Mediated Difunctionalization of Imidazopyridines with Sodium Sulfinates: Synthesis of Sulfones and Sulfides, *J. Org. Chem.*, 2018, **83**, 338–349.

19 C. Breton-Patient, D. Naud-Martin, F. Mahuteau-Betzer and S. Piguel, Three-Component C-H Bond Sulfonylation of Imidazoheterocycles by Visible-Light Organophotoredox Catalysis, *Eur. J. Org. Chem.*, 2020, **2020**, 6653–6660.

20 (*a*) M. Jiang, Y. Yuan, T. Wang, Y. Xiong, J. Li, H. Guo and A. Lei, Exogenous-oxidant- and catalyst-free electrochemical deoxygenative C2 sulfonylation of quinoline N-oxides, *Chem. Commun.*, 2019, **55**, 13852–13855; (*b*) Y.-C. Wu, S.-S. Jiang, S.-Z. Luo, R.-J. Song and J.-H. Li, Transition-metal- and oxidant-free directed anodic C-H sulfonylation of N,N-disubstituted anilines with sulfinates, *Chem. Commun.*, 2019, **55**, 8995–8998; (*c*) Y. Gao, H. Mei, J. Han and Y. Pan, Electrochemical Alkynyl/Alkenyl Migration for the Radical Difunctionalization of Alkenes, *Chem. – Eur. J.*, 2018, **24**, 17205–17209; (*d*) D. Zhang, J. Cai, J. Du, X. Wang, W. He, Z. Yang, C. Liu and Z. Fang, Oxidant- and Catalyst-Free Synthesis of Sulfonated Benzothiophenes via Electrooxidative Tandem Cyclization, *J. Org. Chem.*, 2021, **86**, 2593–2601; (*e*) W. Kim, H. Y. Kim and K. Oh, Electrochemical Radical-Radical Cross-Coupling Approach between Sodium Sulfinates and 2*H*-Indazoles to 3-Sulfonylated 2*H*-Indazoles, *Org. Lett.*, 2020, **22**, 6319–6323; (*f*) Y. Kong, Y. Li, M. Huang, J. K. Kim and Y. Wu, An electrochemical off-on method for pyrimidin-2(1*H*)-one synthesis via three-component cyclization, *Green Chem.*, 2019, **21**, 4495–4498; (*g*) M. Gong, J. K. Kim, X. Zhao, Y. Li, J. Zhang, M. Huang and Y. Wu, Visible-light-induced α-oxyamination of 1,3-dicarbonyls with TEMPO via a photo (electro)catalytic process applying a DSSC anode or in a DSSC system, *Green Chem.*, 2019, **21**, 3615–3620; (*h*) Y. Kong, J. K. Kim, Y. Li, M. Huang, J. Zhang and Y. Wu, An Oxidant- and Catalyst-Free Electrooxidative Cross-Coupling Approach to 3-Tetrahydroisoquinoline Substituted Coumarins, *Green Chem.*, 2021, **23**, 1274–1279.

21 J. Li, L. He, X. Liu, X. Cheng and G. Li, Electrochemical Hydrogenation with Gaseous Ammonia, *Angew. Chem.*, 2019, **58**, 1759–1763.

22 (*a*) E. Laviron, Adsorption, autoinhibition and autocatalysis in polarography and in linear potential sweep voltammetry, *J. Electroanal. Chem.*, 1974, **52**, 355–393; (*b*) P. Qian, M. Bi, J. Su, Z. Zha and Z. Wang, Electrosynthesis of (*E*)-Vinyl Sulfones Directly from Cinnamic Acids and Sodium Sulfinates via Decarboxylative Sulfono Functionalization, *J. Org. Chem.*, 2016, **81**, 4876–4882; (*c*) M.-J. Luo, B. Liu, Y. Li, M. Hu and J.-H. Li, Electrochemical Three-Component 1,2-Aminosulfonylation of Alkenes: Entry to 2-sulfonylethan-1-amines, *Adv. Synth. Catal.*, 2019, **361**, 1538–1542.

23 (*a*) L. Xing and C. Li, Stereoselective Chlorination and Bromination of Enamides and Enamines via an Electrostatic Attraction Effect Using (1,1-Diacetoxyiodo) benzene and a Halide Source, *J. Org. Chem.*, 2015, **80**, 10000–10008; (*b*) S. O. C. Mundle, G. W. Howe and R. Kluger, Origins of Steric Effects in General-Base-Catalyzed Enolization: Solvation and Electrostatic Attraction, *J. Am. Chem. Soc.*, 2012, **134**, 1066–1070.

24 (*a*) A. D. Becke, Density-functional thermochemistry. III. The role of exact exchang, *J. Chem. Phys.*, 1993, **98**, 5648–5652; (*b*) M. Cossi, V. Barone, R. Cammi and J. Tomasi, Ab initio study of solvated molecules: a new implementation of the polarizable continuum model, *Chem. Phys. Lett.*, 1996, **255**, 327–335; (*c*) J. Tomasi, R. Cammi, B. Mennucci, C. Cappelli and S. Corni, Molecular properties in solution described with a continuum solvation model, *Phys. Chem. Chem. Phys.*, 2002, **4**, 5697–5712; (*d*) Y. Zhao and D. G. Truhlar, Density Functionals with Broad Applicability in Chemistry, *Acc. Chem. Res.*, 2008, **41**, 157–167; (*e*) S. Grimme, S. Ehrlich and L. Goerigk, Effect of the Damping Function in Dispersion Corrected Density Functional Theory, *J. Comput. Chem.*, 2011, **32**, 1456–1465.

25 (*a*) P. Bhasi, Z. P. Nhlabatsi and S. Sitha, Expanding the applicability of electrostatic potentials to the realm of transition states, *Phys. Chem. Chem. Phys.*, 2016, **18**, 13002–13009; (*b*) J. S. Murray and P. Politzer, The electrostatic potential: an overview, *Wiley Interdiscip. Rev.: Comput. Mol. Sci.*, 2011, **1**, 153–163; (*c*) A. Mehmood, S. I. Jones, P. Tao and B. G. Janesko, An Orbital-Overlap Complement to Ligand and Binding Site Electrostatic Potential Maps, *J. Chem. Inf. Model.*, 2018, **58**, 1836–1846.

26 T. Lu and F. Chen, Multiwfn: A Multifunctional Wavefunction Analyzer, *J. Comput. Chem.*, 2012, **33**, 580–592.

27 W. Humphrey, A. Dalke and K. Schulten, VMD: Visual Molecular Dynamics, *J. Mol. Graphics*, 1996, **14**, 33–38.

28 N. K. Nandwana, S. Dhiman, V. N. Shinde, U. Beifuss and A. Kumar, Synthesis of π-Expanded Azole-Fused Imidazo[1,2-*a*]pyridine Derivatives and their Photophysical Properties, *Eur. J. Org. Chem.*, 2020, **2020**, 2576–2582.

29 (*a*) C. E. Gueiffier, C. Croix, M. Hervet, J.-Y. Kazock, A. Gueiffier and M. Abarbri, Convenient Synthesis of Alkenyl-, Alkynyl-, and Allenyl-Substituted Imidazo[1,2-*a*]pyridines via Palladium-Catalyzed Cross-Coupling Reactions, *Helv. Chim. Acta*, 2007, **90**, 2349–2367; (*b*) H. Kong, Q. Li, Y. Yin, M. Huang, J. K. Kim, Y. Zhu, Y. Li and Y. Wu, An efficient light on-off one-pot method for the synthesis of 3-styryl coumarins from aryl alkynoates, *Org. Biomol. Chem.*, 2019, **17**, 4621–4628; (*c*) X. Wu, M. Jia, M. Huang, J. K. Kim, Z. Zhao, J. Liu, J. Xi, Y. Li and Y. Wu, A visible-light-induced "on-off" one-pot synthesis of 3-arylacetylene coumarins with AIE properties, *Org. Biomol. Chem.*, 2020, **18**, 3346–3353.